

INTRODUCTION TO LARGE SCALE DATA ANALYTICS

- 1. Classification of Digital Data, Structured and Unstructured Data**
- 2. Introduction to Big Data: Characteristics – Evolution – Definition**
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INTRODUCTION:

LARGE SCALE data analytics is a field of study and practice that focuses on extracting valuable insights and meaningful patterns from large and complex datasets. It involves the use of various techniques, tools, and technologies to process, analyze, and interpret massive volumes of data to make data-driven decisions, identify trends, and gaining valuable insights.

The term "**big data**" refers to the vast amounts of structured, semi-structured, and unstructured data that organizations and businesses collect from various sources such as social media, sensors, mobile devices, transaction records, and more. This data is typically characterized by its volume, velocity, variety, and veracity, which makes it challenging to manage and analyze using traditional data processing methods.

Big data analytics encompasses several key components and techniques:

1. **Data collection:** Gathering and aggregating data from multiple sources, including structured databases, log files, social media platforms, and IoT devices.
2. **Data storage and management:** Storing and organizing large volumes of data using distributed file systems, NoSQL databases, and data warehouses.
3. **Data preprocessing:** Cleaning, transforming, and filtering the data to ensure its quality, consistency, and relevance for analysis.
4. **Data analysis:** Applying various statistical, machine learning, and data mining techniques to extract patterns, correlations, and insights from the data. This can involve tasks such as data exploration, clustering, classification, regression, and predictive modelling.
5. **Data visualization:** Presenting the analyzed data in a visual format, such as charts, graphs, and dashboards, to facilitate better understanding and decision-making.
6. **Real-time analytics:** Performing analysis on streaming data to gain immediate insights and enable real-time decision-making.
7. **Data security and privacy:** Ensure appropriate measures are in place to protect sensitive data and comply with relevant regulations and privacy policies.

Sources of Big Data:

These data come from many sources like

- **Social networking sites:** Facebook, Google, and LinkedIn generate huge amounts of data on a day-to-day basis as they have billions of users worldwide.
- **E-commerce site:** Sites like Amazon, Flipkart, and Alibaba generate many logs from which users buying trends can be traced.
- **Weather Station:** All the weather station and satellite gives huge amounts of data that are stored and manipulated to forecast weather.
- **Telecom company:** Telecom giants like Airtel, and Vodafone study user trends and accordingly publish their plans; for this, they store the data of its million users.
- **Share Market:** Stock exchange worldwide generates huge amounts of data through daily transactions.

Big data analytics:

Big data analytics is the process of extracting meaningful insights from big data, such as hidden patterns, unknown correlation, market trends and customer preferences, that can help organizations make informed business decisions.

There are quite a few advantages to incorporating big data analytics into a business or organization. These include:

- **Cost reduction:** big data can reduce costs in storing all the business data in one place. Tracking analytics also helps companies find ways to work more efficiently to cut costs wherever possible.
- **Product development:** Developing and marketing new products, services, or brands is much easier when based on data collected from customers' needs and wants. Big data analytics also helps businesses understand product viability and keep up with trends.

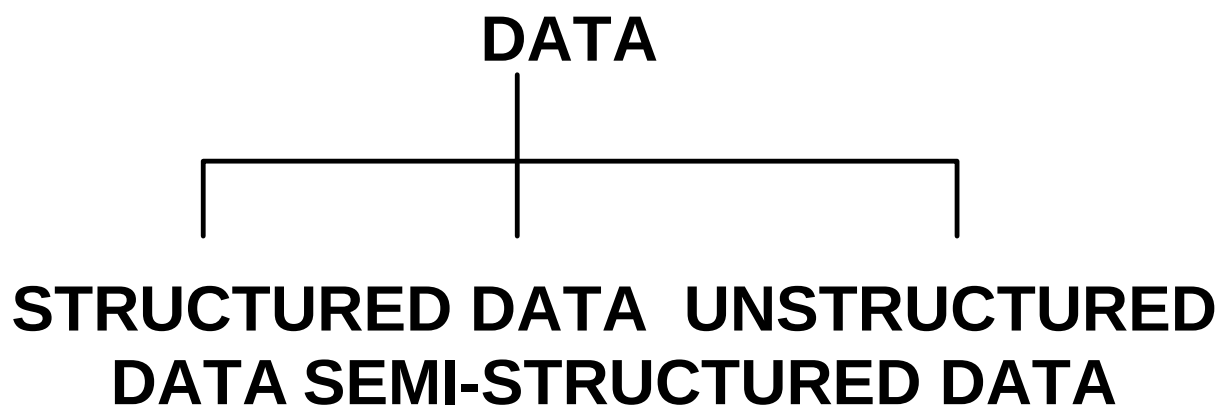
- **Strategic business decisions:** The ability to constantly analyze data helps businesses make better and faster decisions, such as cost and supply chain optimization.
- **Customer experience:** Data-driven algorithms help marketing efforts (targeted ads, for example) increase customer satisfaction by delivering an enhanced customer experience.
- **Risk management:** Businesses can identify risks by analyzing data patterns and developing solutions for managing those risks.

I .Data Classification:

Process of classifying data in relevant categories to be used or applied more efficiently. The classification of data makes it easy for the user to retrieve it. Data classification holds its importance regarding data security and compliance and meeting different business or personal objectives. It is also a major requirement, as data must be easily retrievable within a specific period.

Types of Data Classification:

Data can be broadly classified into 3 types.



1. Structured Data:

- Structured data is created using a fixed schema and is maintained in tabular format. The elements in structured data are addressable for effective analysis.
- It contains all the data which can be stored in the SQL database in a tabular format.

- Today, most data is developed and processed in the simplest way to manage information.

Relational data, Geo-location, credit card numbers, addresses, etc.

- Consider an example of Relational Data you have to maintain a record of students for a university, like a student's name, ID, address, and Email of the student. To store the record of students, the following relational schema and table were used.

S_ID	S_Name	S_Address	S_Email
1001	A	Delhi	A@gmail.com
1002	B	Mumbai	B@gmail.com

2. Unstructured Data:

- It is defined as the data that **does not follow a pre-defined standard, or** you can say that any does not follow any organized format.
- This kind of data is **also not fit for the relational database because** you will see a predefined manner or organized way of data in the relational database.
- Unstructured data is also very important for the big data domain and to manage and store Unstructured data there are many platforms to **handle it like No-SQL Database.**
Examples – Word, PDF, text, media logs, etc.

3. Semi-Structured Data:

Semi-structured data is information that does not reside in a relational **database but that have some organizational properties that make it easier to analyze.** With some process, you can store them in a [relational database](#) but is very hard for some semi-structured data, but semi-structured exist to ease space.

Example – [XML data](#).

Features of Data Classification:

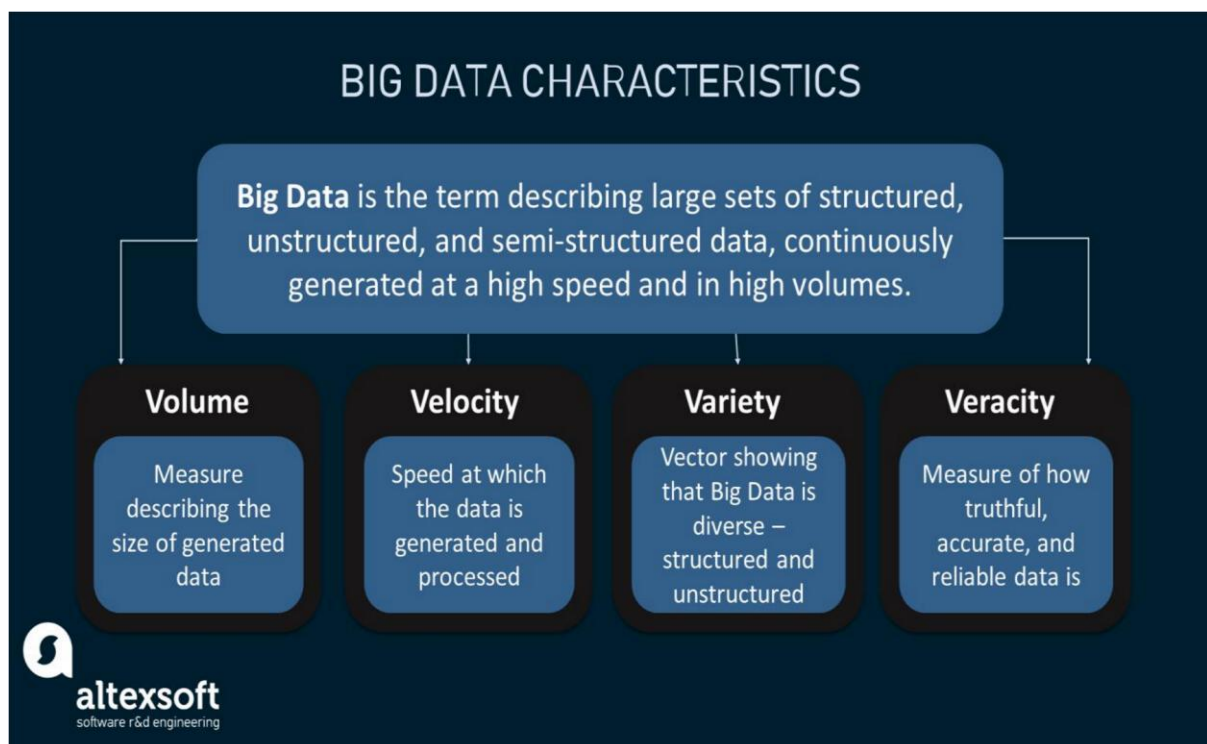
The main goal of the organization of data is to arrange the data in such a form that it becomes fairly available to the users. So, it's basic features as following.

- **Homogeneity** – The data items in a particular group should be similar to each other.
- **Clarity** – There must be no confusion in positioning any data item in a particular group.
- **Stability** – The data item set must be stable i.e. any investigation should not affect the same set of classification.
- **Elastic** – One should be able to change the basis of classification as the purpose of classification changes.

II. Characteristics of Big Data

Big Data contains much data not being processed by traditional data storage or the processing unit. Many multinational companies use it to **process** the data and business of many **organizations**. The data flow would exceed **150 exabytes** per day before replication.

There are five v's of Big Data that explains the characteristics.



5 V's of Big Data

- Volume
- Veracity ○ Variety ○ Value ○ Velocity

- **1. Volume**
- The name Big Data itself is related to its enormous size. Big Data is a vast ‘volume’ of data generated from many sources daily, such as business processes, machines, social media platforms, networks, human interactions, and many more.

- Facebook can generate approximately a billion messages, 4.5 billion times that the "Like" button is recorded, and more than 350 million new posts are uploaded daily. Big data technologies can handle large amounts of data.

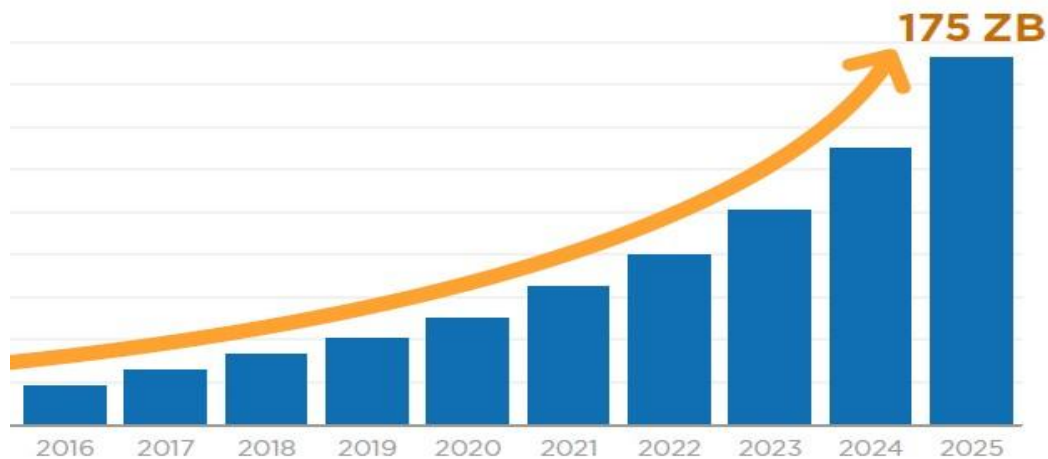


Fig: Expected volume of Big data

2. Variety

Big Data can be structured, unstructured, semi-structured and collected from **different sources**. Data will only be collected from databases and sheets in the past, But these days the data will come in array forms, PDFs, Emails, audio, SM posts, photos, videos, etc.

3. Veracity

Veracity **means how much the data is reliable**. It has many ways to filter or translate the data. Veracity is the **process of being able to handle and manage data efficiently**. Big Data is also essential in business development.

For example, Facebook posts with hashtags.

4. Value

Value is an essential characteristic of big data. It is not the data that we process or store. It is **valuable and reliable data that we store, process, and also analyze.**

5. Velocity

Velocity plays an important role compared to others. Velocity creates the speed by which the **data is created in real-time. It contains the linking of incoming data set speeds, rate of change, and activity bursts.** The primary aspect of Big Data is to provide demanding data rapidly.

Big data velocity deals with the speed of data flows from sources like application logs, business processes, networks, social media sites, sensors, mobile devices, etc.

○ Every one minute

100000 + tweets ,

69500 + status updates ,

11000000 +internet messages ,

698445 +google searches ,

168000000 + emails ,

1820 TB data created

III. History of big data / Evolution of big data

The history of big data can be traced back to the early days of computing and the emergence of digital data. Here's a brief overview of the key milestones in the evolution of big data:

1. **Emergence of Databases:** In the 1960s, the concept of databases and data management systems began to take shape. Developing relational database management systems (RDBMS) like IBM's System R and Oracle's RDBMS laid the foundation for structured data storage and retrieval.
2. **Data Warehousing:** In the 1980s, the concept of data warehousing emerged. Data warehousing involves consolidating data from multiple sources into a central repository for analysis and reporting purposes. This approach enabled organizations to leverage their data assets for decision-making.
3. **Internet and Web 1.0:** The rise and the advent of the World Wide Web in the 1990s brought about a massive increase in data generation. The web became a platform for publishing, sharing, and collecting data, leading to the growth of unstructured and semistructured data sources.
4. **Digital Transformation and Enterprise Systems:** Organizations started generating vast amounts of structured data with the widespread adoption of enterprise systems such as Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM). This marked the beginning of data-driven decision-making within businesses.
5. **Hadoop and Distributed Computing:** In 2005, Doug Cutting and Mike Cafarella developed Hadoop, an open-source framework designed to process large datasets in a distributed computing environment. Hadoop's distributed file system (HDFS) and MapReduce programming model made it easier to handle big data by breaking it into smaller parts and processing them in parallel.
6. **The Rise of NoSQL:** Traditional relational databases struggled to handle big data's volume, variety, and velocity. Consequently, the development of NoSQL (Not Only SQL) databases gained prominence in the late 2000s. NoSQL databases offered flexible schema designs and horizontal scalability, making them suitable for handling unstructured and semi-structured data.
7. **Advanced Analytics and Machine Learning:** As big data grew, organizations sought to extract valuable insights and patterns from the data. Advanced analytics techniques, such as data mining, machine learning, and predictive modeling, gained traction for uncovering actionable insights and supporting decision-making.

8. **Cloud Computing:** The advent of cloud computing in the late 2000s provided organizations with scalable and cost-effective infrastructure for storing, processing, and analyzing big data. Cloud platforms, such as Amazon Web Services (AWS) and Google Cloud Platform (GCP), offered on-demand resources, eliminating the need for largescale on-premises infrastructure.
9. **Internet of Things (IoT):** With the proliferation of connected devices and sensors, the amount of data generated by IoT devices skyrocketed. The IoT contributed to the expansion of big data, as real-time streaming data from various sources became available for analysis.
10. **Current Trends:** Today, big data analytics continues to evolve, driven by advancements in technologies such as artificial intelligence, natural language processing, and deep learning. The focus has shifted towards real-time analytics, data governance, privacy concerns, and the ethical use of data.

The history of big data demonstrates the increasing importance of managing and analyzing large and diverse datasets. The field continues to evolve, and organizations are continually exploring innovative approaches and technologies to leverage the potential of big data for insights and decision-making.

IV. Challenges of Big Data

The challenges in **Big Data** are the real implementation hurdles. These require immediate attention and need to be handled because if not handled then the failure of the technology may take place which can also lead to some unpleasant result. Big data challenges include the storing, analyzing the extremely large and fast-growing data.

1. Incomplete Understanding of Big Data:

- Sometimes organizations are unable to understand what is data? Why is it so important? because of this integration of required algorithms to analyze the data becomes really difficult.
- Skilled persons required.

- For example, if employees do not understand the importance of data storage, they might not keep the backup of sensitive data. They might not use databases properly for storage. As a result, when this important data is required, it cannot be retrieved easily

2. Exponential Data Growth:

- One of the biggest challenges is the exponential growth of data as time goes on.
- Since most of the data is unstructured and comes in the form of the document, videos, etc. Which is difficult to analyze.

Since data is growing exponentially, it's becoming difficult to store this data.

3. Security of Data

- Companies spend so many resources for understanding, storing, and analyzing the data they often forget to prioritize the security part.
- So, they try to push the security part for the later stages because of which data breach is possible.

- A solution for these companies is hiring cybersecurity professionals to protect their data.

4. Data Integration

- Since data can be in any form from structured data like phone numbers to unstructured data like videos. So, integrating these data is difficult, but at the same time, data integration is crucial for processes like analysis, reporting, and forecasting.
- So integration has to be error-free. Companies can solve these problems by purchasing the right technologies like IBM Infosphere, Microsoft SQL, etc.

5. Confusion while Big Data tool selection

- Companies often get confused while selecting the best tool for Big Data analysis and storage. Is HBase or Cassandra the best technology for data storage? Is Hadoop MapReduce good enough or will Spark be a better option for data analytics and storage?
- These questions bother companies and sometimes they are unable to find the answers. They end up making poor decisions and selecting inappropriate technology. As a result, money, time, efforts and work hours are wasted.

6. Lack of data professionals

○ To run these modern technologies and Big Data tools, companies need skilled data professionals. These professionals will include data scientists, data analysts and data engineers who are experienced in working with the tools and making sense out of huge data sets.

VII. Classification of Data analytics

Big data analytics is process to extract meaningful insights from big data such as hidden patterns, unknown correlation, market trends and customer preferences that can help organizations make informed business decisions.

- Data analytics can be of four types depending on the type and scope of analysis being conducted on the data set.

1. Descriptive analytics

2. Diagnostic analytics

3. Predictive analytics

4. Prescriptive analytics

Descriptive analytics: What happened

- Descriptive analytics uses historical data from a single internal source to describe what happened.
- For example: How many people viewed the website?
- Which products had the most defects?
- Used by most businesses, descriptive analytics forms the crux of everyday reporting, especially through dashboards.

Diagnostic analytics: Why did it happen?

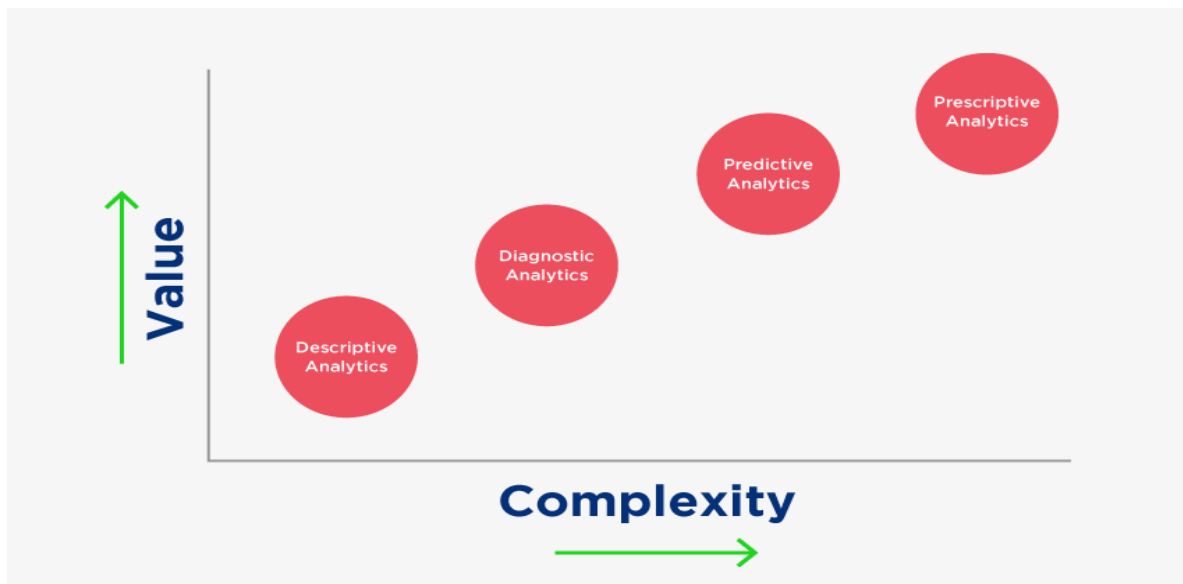
- Diagnostic analytics is a form that dives deep into historical data to **identify anomalies, find patterns, identify correlations, and determine causal relationships.**
- With the help of diagnostic analysis, data analysts **can understand why a certain product did not do well in the market, or why customer satisfaction decreased in a certain month.**

Predictive analytics: What might happen next?

- **This is a more advanced form of analytics that is often used to answer the question ‘what will happen next?’ in a business situation.**
- As the name suggests, this data analytics type predicts the future outcome of a situation depending on all the available data.
- This data will include both market trends and older data about your business performance, By combining these two, this data analytics type can predict how your business will perform during the next season

Prescriptive analytics: What do I need to do?

- Prescriptive analytics is the most complex type of analytics.
- It combines internal data, external sources, and machine-learning techniques to provide the most effective recommendations for business decisions.



VIII. Big data analytics challenges

1. Data quality :

- Poor data quality can significantly impact the accuracy and reliability of analytical results. Incomplete, inconsistent, and inaccurate data can lead to biased or misleading insights.

2. Data Integration:

- Organizations often have data stored in different systems and formats across various departments or sources.
- Integrating data from disparate sources can be complex and time-consuming. Data integration challenges include data harmonization, resolving schema and format differences, and aligning data from different databases or systems..

3. Data Privacy and Security:

- With the increasing focus on data privacy regulations, protecting sensitive and personal data is a significant challenge.

4. Skill Gap and Talent Acquisition:

- There is a shortage of skilled data analysts, data scientists, and data engineers who possess the necessary expertise in data analytics.
- Organizations may struggle to find and retain professionals with the right skill set to drive effective data analytics initiatives.

5. Technology Selection and Integration:

- The data analytics landscape is vast, with numerous tools, platforms, and technologies available.
- Selecting the right technologies that align with organizational needs and integrating them seamlessly with existing systems can be challenging.

6. Scalability and Performance:

- As data volumes and complexity grow, organizations must ensure that their analytics infrastructure can scale accordingly.
- Processing and analyzing large datasets within acceptable timeframes can strain computational resources and affect performance. Designing scalable architectures and optimizing query performance are vital.

7. Technology Selection and Integration:

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8. Cost Management:

- Implementing and maintaining a robust data analytics infrastructure can be costly.
- Investing in hardware, software, talent, and ongoing maintenance and support can strain budgets.

IX. Top analytical tools

There are several top analytical tools available in the market that are widely used for processing, analyzing, and deriving insights from data. Here are some of the popular analytical tools.

Python:

- Python is a versatile programming language widely used for data analysis and scientific computing.
- It offers numerous libraries and frameworks, such as pandas, NumPy, and scikit-learn, which provide powerful data manipulation, analysis, and machine learning capabilities.

- **R:**

- R is a programming language and environment specifically designed for statistical computing and graphics.
- It offers a wide range of packages and libraries for data manipulation, analysis, and visualization.
- R is popular among statisticians and data scientists for its extensive statistical modeling capabilities

Apache Spark:

- Apache Spark is an open-source distributed computing system that provides fast and scalable data processing and analytics capabilities.

- It supports various programming languages and offers libraries for distributed data processing, machine learning, and graph analytics.

Apache Hadoop: Apache Hadoop is an open-source framework for distributed processing of large datasets.

- It provides a scalable and fault-tolerant platform for storing, processing, and analyzing big data.
- Hadoop includes components such as Hadoop Distributed File System (HDFS) and MapReduce for distributed data processing.

Cassandra

- [APACHE Cassandra](#) is an open-source NoSQL distributed database that is used to fetch large amounts of data.
- It is capable of delivering thousands of operations every second and can handle petabytes of resources with almost zero downtime.

Mongo DB

- It is a free, open-source platform and a document-oriented (NoSQL) database that is used to store a high volume of data.
- It uses collections and documents for storage and its document consists of key-value pairs which are considered a basic unit of [Mongo DB](#).

It is so popular among developers due to its availability for multi-programming languages such as Python, Jscript, and Ruby

Tableau:

- Tableau is a powerful and user-friendly data visualization and business intelligence tool.

- It allows users to create interactive dashboards, reports, and visualizations from various data sources. Tableau supports advanced analytics, data blending, and provides intuitive drag-and-drop functionality.

Microsoft Power BI:

- Microsoft Power BI is a business intelligence tool that helps users analyze data and share insights.
- It provides interactive dashboards, data visualization, and reporting capabilities.
- Power BI integrates with various data sources, offers AI-powered features, and supports collaboration and data sharing.

SAS:

- SAS (Statistical Analysis System) is a comprehensive suite of analytics tools widely used in industries such as finance, healthcare, and government.
- It provides a wide range of functionalities for data management, advanced analytics, and reporting.

QlikView/Qlik Sense:

- QlikView and Qlik Sense are data visualization and business intelligence tools that enable users to explore and analyze data visually.
- They offer drag-and-drop functionality, interactive dashboards, and powerful data discovery capabilities.

MATLAB:

- MATLAB is a programming and numerical computing environment widely used in academia and industry.
- It offers comprehensive tools for data analysis, modeling, and simulation, along with built-in algorithms and functions for various analytical tasks.

- WEKA

WEKA stands for **Waikato Environment for Knowledge Analysis**.

It is a **popular open-source machine learning and data mining tool** developed at the University of Waikato, New Zealand.

Here's a breakdown for you:

◆ Key Features of WEKA

- **GUI-based:** Provides a graphical interface for easy use (no need for complex coding).
- **Data Preprocessing:** Includes tools for cleaning, transforming, and filtering data.
- **Machine Learning Algorithms:** Supports classification, regression, clustering, association rule mining, and feature selection.
- **Visualization:** Offers graphs and charts to visualize datasets and model outputs.
- **Extensible:** Written in Java and supports plugins, making it customizable.

◆ Applications

- Educational purposes (teaching data science & ML concepts).
- Research in data mining and machine learning.
- Practical tasks such as:
 - Text mining
 - Fraud detection
 - Bioinformatics
 - Market basket analysis

◆ Example Algorithms Available in WEKA

- **Classification:** Decision Trees (J48), Naive Bayes, SVM.
- **Clustering:** K-means, EM.
- **Regression:** Linear Regression, M5P.
- **Association:** Apriori for market basket analysis.

☞ In short: WEKA is a **beginner-friendly yet powerful machine learning toolkit**, widely used for experimentation, learning, and rapid prototyping.

Do you want me to **add WEKA into your converted Word notes** under the *Analytical Tools* section (along with Python, R, Spark, etc.)?

IMPORTANT QUESTIONS

1. What is Big Data ? Explain Characteristics of Big data?
2. What is big data analytics? Explain five 'V's of Big data. Briefly discuss applications of big data?
3. What are the benefits of Big Data? Discuss challenges under Big Data.?
4. Explain the difference between structure and unstructured data.
5. What is big data analytics and discuss the different types of big data analytics?
6. Differentiate between traditional BI and Big data?
7. Discuss about data ware housing and Hadoop frame work?